SARASWAT ACADEMY

Class Test - Chapter 4: Describing Motion Around Us

Grade: 9 | Time: 1 Hour | Total Marks: 20

Instructions:

• Answer all questions. • Write legibly and clearly. • Show all calculations and reasoning. • Use appropriate units in your answers.

Section A: Multiple Choice Questions (5 marks)

1. An object travels 100 m in 20 seconds. What is its average speed?

(a) 5 m/s (b) 10 m/s (c) 20 m/s (d) 200 m/s

2. The difference between distance and displacement is that:

(a) Distance depends on path; displacement does not

(b) Distance is a scalar; displacement is a vector

(c) Both (a) and (b)

(d) Distance is always greater than displacement

3. A car accelerates at 2 m/s^2. Starting from rest, what is its velocity after 5 seconds?

(a) 2 m/s (b) 5 m/s (c) 10 m/s (d) 25 m/s

4. In a position-time graph, a straight line indicates:

(a) Acceleration (b) Constant velocity (c) Variable velocity (d) At rest

5. In uniform circular motion, velocity is constant but acceleration is not zero because:

(a) Speed changes (b) Direction changes (c) Both speed and direction change (d) It moves in a straight line

Section B: Assertion-Reason Questions (2 marks)

Direction: In the question given below, there are two statements - Assertion (A) and Reason (R). Select the correct option:

(i) Both A and R are true, and R is the correct explanation of A.

(ii) Both A and R are true, but R is not the correct explanation of A.

(iii) A is true, but R is false.

(iv) A is false, but R is true.

Assertion (A): A car moving at constant velocity on a straight highway has zero acceleration.

Reason (R): Acceleration is the rate of change of velocity, and when velocity is constant, there is no change in velocity.

Section C: Case Study Question (4 marks)

Sarang is riding his bicycle along a straight road. He starts from rest at point O and accelerates uniformly at 2 m/s^2 for the first 10 seconds. After this, he maintains a constant velocity for the next 15 seconds. Then, he applies the brakes and comes to rest in the next 5 seconds with uniform deceleration.

Based on this information, answer the following:

(a) Calculate the maximum velocity Sarang attains during his ride. (1 mark)

(b) Find the total distance covered by Sarang during the entire journey. (1.5 marks)

(c) Calculate the deceleration when brakes are applied. (1.5 marks)

Section D: Short Answer Questions (5 marks)

6. A student walks 10 m north, then 10 m south, returning to the starting point. Calculate the distance travelled and displacement. (1.5 marks)

7. What is the significance of slope in a velocity-time graph? (1.5 marks)

8. How can the area under a velocity-time curve represent displacement? Explain briefly. (2 marks)

Section E: Long Answer Questions (4 marks)

9. Explain the difference between average velocity and instantaneous velocity. Give examples to illustrate why instantaneous velocity becomes important when an object is accelerating. Also, discuss how the velocity-time graph helps distinguish between constant and variable acceleration. (4 marks)